

Detailed

Complete Beginner's Guide: Mi Band 5 → Custom App

What you need before starting:

- A Mi Band 5 (the physical device, charged)
- A Windows or Mac computer (this guide assumes **Mac**, since that's what was used to build/test this process — if you're on Windows, most steps are the same but the Terminal commands look slightly different, flag it and ask)
- An Android phone (this guide assumes Android, since some tools used later only work on Android, not iPhone)
- A Google account (to use the Play Store)
- An email address you're okay creating a new account with (Gmail is fine)

What you'll end up with by the end: Your own custom app on your phone showing live heart rate, step count, and sleep data pulled directly from the Mi Band 5.

GLOSSARY — words you'll see below, explained simply

- **App Store / Play Store:** the app on your phone icon (usually a colorful triangle/bag icon) where you download apps. On Android it's called "Play Store" or "Google Play."
- **F-Droid:** a *second*, separate app store (not Google's) that has open-source apps not found on the Play Store. You'll install this one manually.
- **Terminal:** a black/white text window on your Mac where you type commands instead of clicking buttons. It looks scary but you'll just copy-paste exact text.
- **Bluetooth / BLE:** how your phone talks to the watch wirelessly, short-range.

- **Auth key / pairing key:** a secret code, like a password, that lets an app talk to your specific band. Without it, apps other than Zepp Life are blocked from reading the band's data.
 - **GATT / BLE characteristic:** technical Bluetooth terms for "a specific piece of data the band can send or receive" (e.g., one characteristic = heart rate, another = battery level). You don't need to memorize this, just recognize it later in code.
 - **Android Studio:** the free program used to build Android apps.
 - **APK:** the actual installable Android app file, like a `.exe` on Windows.
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PART A — Get the band talking to your phone (official way first)

A1. Charge the band

Plug the Mi Band 5 into its charging cradle/cable, plug that into a USB port or wall adapter, leave it until it shows a full battery icon on the band's screen (usually 1-2 hours).

A2. Install Zepp Life on your Android phone

1. On your phone, find the **Play Store** app icon (colorful triangle) and tap it
2. Tap the search bar at the top
3. Type: `Zepp Life`
4. Tap the first result (icon is usually orange/red with a stylized "Z" or similar)
5. Tap **Install**
6. Wait for it to download and install, then tap **Open**

A3. Create an account in Zepp Life

1. When the app opens, it'll ask you to sign up or log in
2. **Important — write this down for later:** choose "Sign up with email," not "Sign in with Google" — this matters for a step later on. Use a real email you have

access to.

3. Enter an email address and create a password. **Write both down somewhere safe** — you'll need them again in Part B.
4. Confirm/verify the email if it asks (check your inbox for a confirmation code or link)

A4. Pair the Mi Band 5

1. In the Zepp Life app, look for a "+" button or "Add Device" option (usually on the main/home screen)
2. Select **Mi Band 5** from the device list shown
3. Make sure Bluetooth is turned on on your phone (swipe down from the top of the screen, tap the Bluetooth icon if it's not already blue/on)
4. Follow the on-screen pairing instructions — it'll ask you to confirm a code shown on the band, or just tap "pair" when it appears
5. Wait for it to say "Connected" or "Sync complete"

Checkpoint: you should now see your steps/battery showing up in the Zepp Life app. If yes, move to Part B.

PART B — Extract the secret key (on your Mac)

This step gets the "auth key" — a secret code needed so that *other* apps (not just Zepp Life) can read your band's data.

B1. Open Terminal

1. Press **Cmd** + **Space** together on your keyboard (this opens Spotlight search)
2. Type: **Terminal**
3. Press Enter
4. A black/white window opens — this is Terminal. Don't be scared, you're just going to copy-paste text into it.

B2. Check if git is installed

1. In Terminal, type exactly: `git --version` and press Enter
2. If a version number shows up (like `git version 2.39.0`), skip to B3
3. If instead a popup appears asking to install "Command Line Developer Tools," click **Install**, agree to the license, and wait a few minutes for it to finish

B3. Download the key-extraction tool

1. In Terminal, copy and paste this exactly, then press Enter:

```
git clone <https://github.com/argrento/huami-token.git>
```

2. You'll see some text scroll by ending in something like "done." — that means it worked.

B4. Move into the downloaded folder

1. Type: `cd huami-token` and press Enter
2. (This just tells Terminal "go inside that folder you just downloaded")

B5. Create a safe, isolated Python environment

1. Type: `python3 -m venv venv` and press Enter. Wait a few seconds.
2. Type: `source venv/bin/activate` and press Enter
3. Your Terminal line should now start with `(venv)` — that confirms it worked

B6. Install the tool's requirements

1. Type: `pip install .` (note: there's a **period/dot** at the very end — don't forget it) and press Enter
2. Wait for it to finish (a few lines of text, ending without a red "ERROR")

B7. Run the tool to get your key

1. Type this, but replace `youremail@example.com` with the real email you used in Zepp Life, and replace `yourpassword` with your real password (keep the single quotes around the password exactly as shown):

```
python3 main.py --method amazfit --email youremail@example.com --password 'yourpassword' --bt_keys
```

2. Press Enter
3. **Do NOT paste this command with your real email/password anywhere else, including into chat with anyone helping you** — type it directly into Terminal only
4. After a few seconds, look for a block of text like this:

```
Device 0:  
MAC: XX:XX:XX:XX:XX:XX  
Key: 0XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

5. **Write down the MAC and Key values somewhere safe** — you'll enter these in Part C. These two values are what unlock your band for other apps.

If you get a "401" or login error here: it usually means you signed into Zepp Life using "Sign in with Google" instead of email+password. Go back into the Zepp Life app → your profile/account settings → look for "Account Security" or "Set Password" → add an email+password login option to your existing account (this does NOT remove your Google sign-in or unpair your band) → then redo step B7 with the new password.

PART C — Install Gadgetbridge and confirm the key works

Gadgetbridge is a free, open-source app that proves your key works before we build anything custom.

C1. Turn on "Unknown Sources" installs (needed because F-Droid isn't on the Play Store)

1. On your Android phone, open **Settings**
2. Search for "Install unknown apps" or "Unknown sources" in the Settings search bar

3. Find your phone's web browser app (e.g., Chrome) in the list, tap it, and turn on "Allow from this source"

C2. Download F-Droid

1. Open your phone's web browser (Chrome, etc.)
2. Go to: `f-droid.org`
3. Tap the **Download F-Droid** button
4. Once downloaded, tap the notification/file to open and install it — it'll ask for permission (the "unknown sources" toggle you just enabled), tap **Install**

C3. Install Gadgetbridge through F-Droid

1. Open the **F-Droid** app you just installed
2. Tap the search icon (magnifying glass)
3. Type: `Gadgetbridge`
4. Tap it, then tap **Install**
5. Wait, then **Open**

C4. Add your Mi Band 5 to Gadgetbridge

1. In Gadgetbridge, tap the "+" button (bottom right, usually blue circle)
2. It'll scan for nearby Bluetooth devices — find and tap your **Mi Band 5** in the list
3. When it asks for the **Auth Key**, type/paste in the Key value you wrote down in Part B (starts with `0x`)
4. Wait for it to say "**Connected**"

Checkpoint: tap the heart icon in the row of icons at the top — it should show a live heart rate number after a few seconds. If it does, your key is 100% confirmed working, and everything after this is normal app-building work — no more risk of the key/pairing not working.

PART D — Build your own custom app

This is the actual coding part. This section assumes zero coding experience, so go slowly.

D1. Install Android Studio

1. On your Mac, open a web browser, go to: `developer.android.com/studio`
2. Click the big **Download** button
3. Once downloaded, open the file and drag the Android Studio icon into the Applications folder (like installing any Mac app)
4. Open Android Studio from your Applications folder
5. It'll run a setup wizard on first open — click **Next** through the defaults, let it download the Android SDK (this can take 10-20 minutes and several GB of space, be patient)

D2. Create your first project

1. In Android Studio's welcome screen, click **New Project**
2. Choose **Empty Activity** (a plain, blank starting template), click **Next**
3. Name: type something like `MibandReader`
4. Language: make sure **Kotlin** is selected (not Java)
5. Minimum SDK: leave as default, or pick API 26 if asked
6. Click **Finish** and wait for it to set up (a minute or two)

D3. Turn on Bluetooth permissions

1. On the left side of Android Studio, find and double-click the file `AndroidManifest.xml` (inside `app > manifests`)
2. Find the line that says `<manifest ...>` near the top
3. Right after that line, add these three lines exactly:

```
<uses-permission android:name="android.permission.BLUETOOTH_SCAN" />
<uses-permission android:name="android.permission.BLUETOOTH
```

```
H_CONNECT" />
<uses-permission android:name="android.permission.ACCESS_F
INE_LOCATION" />
```

4. Save the file (Cmd + S)

D4. Write down the band's protocol addresses (needed later in code)

These are fixed values specific to Mi Band 5 — just keep this list handy, you'll paste these into code in D6:

```
Main service:      0000fee0-0000-1000-8000-00805f9b34fb
Battery info:      00000006-0000-3512-2118-0009af100700
Auth service:      0000fee1-0000-1000-8000-00805f9b34fb
Auth characteristic: 00000009-0000-3512-2118-0009af100700
Heart rate service: 0000180d-0000-1000-8000-00805f9b34fb
Heart rate data:    00002a37-0000-1000-8000-00805f9b34fb
Heart rate control: 00002a39-0000-1000-8000-00805f9b34fb
```

D5. Connect to the band (basic Bluetooth code)

Follow Android's own official tutorial titled **"Connect to a GATT Server"** (search that exact phrase on developer.android.com) — it walks through, with full code you can copy, how to scan for a device and open a connection. Use your band's MAC address (from Part B) as the target device.

D6. Implement the "unlock" step (the hard part)

The band won't send real data until your app proves it knows the secret key — same as Gadgetbridge did automatically for you in Part C, now your own code needs to do it.

Don't write this from scratch — open one of these two free, working examples side-by-side with your project and copy the logic:

- Search GitHub for: [miband-5-heart-rate-monitor](#) (by user Jaapp) — smaller, easier to read
- Search Codeberg for: [Freeyourgadget/Gadgetbridge](#) — the full, production version

This step is the trickiest in the whole guide. If stuck here, that's completely normal — slow down and read the reference code line by line rather than guessing.

D7. Get live heart rate showing

Once the unlock step (D6) works:

1. Tell the band to start sending live heart rate using the "Heart rate control" address from D4
2. "Subscribe" to updates on the "Heart rate data" address — this means: tell Android "notify me every time this value changes"
3. Every time a new number comes in, show it on the phone screen

D8. Get steps and sleep data

These aren't "live" like heart rate — instead your app asks the band "give me your stored activity history," similar to how Zepp Life shows your daily totals. The Gadgetbridge source code (same repo as D6) shows the exact commands used for this — look for the file related to syncing activity data.

D9. Basic screen to show it all

Add a few simple text boxes on screen (a `TextView` for heart rate, one for steps, one for sleep) so you can see the numbers update — this proves everything is working before making it look nice.

Troubleshooting Cheat Sheet

Problem	Likely Fix
Terminal says "command not found"	Retype the command carefully, no typos, no extra spaces
"401" error extracting key	Signed in with Google — add email+password login in Zepp Life account settings, retry
Password has symbols and Terminal errors	Wrap password in single quotes: <code>'password'</code>
Gadgetbridge won't connect	Fully close the Zepp Life app first — only one app can hold the Bluetooth connection at a time

Problem	Likely Fix
Custom app can't find the band	Double check Bluetooth is on, and you're using the correct MAC address from Part B
Custom app connects but gets no data	The "unlock" step (D6) likely isn't implemented correctly yet — re-check against the reference code

Keep these three links open in a browser tab while working through Part D:

- github.com/argrento/huami-token (the key tool, already used)
- codeberg.org/Freeyourgadget/Gadgetbridge (protocol reference for D6-D8)
- developer.android.com/studio (Android Studio docs)